CrossyIce

Project Title : CrossyIce

Overview : This project is a 2D grid based puzzle game built in F# using .NET 10 and Raylib_cs. The user controls a character that must reach a target destination by navigating through diverse terrains, such as dry land and slippery ice, while using bombs to break through the environment.

Requirements :

1. The user will see a top-down 2D cell-based grid map rendered in a window. The map will feature distinct visual entities: dry land (white), slippery ice (sky blue), walls, cracked obstacles, the player character, bombs, and a destination tile.
2. The user will see a UI element displaying the number of bombs remaining for the current stage, the number of current stages.
3. The user will control the player by using the WASD keys. Whenever the user presses W, A, S, or D, the player's facing direction changes to match each input. Movement mechanism has dependency on the tile the player moves onto.
4. If the player moves onto a dry land tile, the character moves exactly one cell. If the player moves onto an ice tile, the player slides continuously in the chosen direction. The character stops on the first dry land tile reached while sliding. If the next cell is a wall, cracked obstacle, bomb, or map boundary, the character stops on the last valid cell before it.
5. The user is provided with a limited number of bombs per stage. The user can press the Spacebar to spawn a bomb directly in front of the character. If the player has no bombs remaining, or if the cell directly in front of the player is not an empty dry land or ice tile, bomb is not placed and the bomb count is not consumed.
6. The user can push a bomb by moving into it. If the bomb is pushed onto a dry land tile, the bomb moves exactly one cell and remains without exploding. If the bomb is pushed onto an ice tile, the bomb slides continuously in that direction. When the user successfully pushes a bomb, the player moves into the bomb's previous cell.
7. If a bomb slides on ice and reaches a dry land tile, it stops on that dry land tile without exploding. If a bomb slides on ice and the next cell is a wall, cracked obstacle, or map boundary, the bomb stops on the last valid ice tile before the blocked cell and explodes there.
8. When a bomb explodes, the blast of cross pattern (the center cell of the explosion plus the four immediately adjacent orthogonal cells) destroys any cracked obstacles in area. Walls, dry land, ice, destination tiles, and other non-cracked tiles remain unaffected. If the player character is caught in the blast area, the stage displays a "Restart?" Screen.

User can restart the current stage by clicking "Restart" button or close the game window by clicking "EXIT" button.

9. The user can press the 'R' key at any time to restart the current stage if they are stuck or out of bombs. Restarting a stage restores the original map layout, player position, bomb count, and removes all placed bombs.
10. The user must navigate from the designated starting coordinate to the destination tile to clear a stage.
11. The game will sequence through 5 different stages. Upon clearing the 5th stage, the game will display a "Game Clear" screen. After the user clicks the "EXIT" button on the "Game Clear" screen, the game window closes.

Example Interaction :

The game renders a level consisting of white dry land tiles, sky blue ice tiles, obstacles, and a goal. The UI shows "Stage : 1", "{Bombs Picture}: 1". The user presses D to change the character’s direction to the right and move one grid cell. The user presses Spacebar to place a bomb, then presses D to push the bomb. The bomb slides into a cracked obstacle, explodes, and clears a path. The user then safely navigates through the newly opened path to reach the goal, and the game automatically loads the next stage.